Your solution will be evaluated based on the following rubrics.

Criteria	Meet Expectations	Does Not Meet Expectations
Task 1: **Data Cleaning and Pre-Processing** **(10%)**	Performed all data quality checks and addressed all data quality issues in the right way (especially the missing value treatment). Provided a clear explanation for missing value removal or imputation	Did not handle the missing values efficiently
	Provided a well-commented explanation for each step of data processing	Did not perform and explain the necessary steps for [illegible] processing in the commen[illegible]
	Dropped the variables that were not relevant for the project with a well-commented explanation for each step. Converted all variables to correct	Did not remove unwanted variables from the columns and did not convert the variables to

Study Plan

Evaluation Rubric

Task 2: **Text Processing** **(10%)**	after cleaning the data set. Provided well-commented reasons for every step that was performed in text preprocessing.	preprocessing steps to clean the text of the reviews. Did not get clean text at the end of this step
Task 3: **Feature Extraction** **(10 %)**	Divided the data into training and testing parts Converted the text to features using the best-suited vectorizer (bag-of-words, TD-IDF, Word2Vec, etc.) for the ML model for sentiment analysis model building.	Did not divide the data into training and testing parts Did not create the appropriate vectorizer for feature extraction to build the ML model
Task 4: **Model Building** **(20%)**	Built at least 3 ML models and did the comparative analysis on why one model is better than the other three models Checked whether the data is imbalanced or not and took the necessary steps. If necessary then did the hyperparameter tuning also Selected one out of the 3 models based on performance for predicting the sentiments based on the text and title of the reviews. Provided detailed	Did not build the 3 ML models and did not give a satisfactory explanation on why one model is better than the other three models Did not consider the imbalance in the data set Did not select the ML model that performed better than the rest. Did not give reasons for

☰ Study Plan

Evaluation Rubric

	recommendation system	train and test datasets
Task 5: Building the Recommendation System (20%)	Built at least two types of recommendation systems: user-based and item-based recommendation systems	Did not build the two types of recommendation systems
	Evaluated both the types of recommendation systems and selected one based on performance. Provided detailed reasons for selecting the recommendation system	Did not evaluate the recommendation systems and did not provide any explanation for selecting the best-suited recommendation system
Task 6: Recommendation of Top 20 Products to a Specified User (10%)	Recommended the top 20 products for the username selected by the user based on the recommendation system built	Did not recommend the top 20 products for any selected username using the finalised recommendation system
Task 7: Fine-Tuning the Recommendation	Predicted the sentiment (positive or negative) of all the reviews in the train data set of the top 20 recommended products for a user. For each of	Did not predict the sentiment using the ML model of all the reviews of the top 20

Evaluation Rubric

Top 5 Products (10%)	of each product. Filtered out the top 5 products with the highest percentage of positive reviews	the percentage of positive reviews
Task 8: Deployment Using Flask (10%)	Build an end-to-end web application using Flask and deploy it in your local system. You need to only include one ML model and one recommendation system to deploy the model. You need to: 1. Take the username as input, and 2. Create a submit button, and once you press the submit button, it should recommend the 5 products based on the username entered.	Was not able deploy the project using Flask